4. Classifiation

Chris Reudenbach

Tasks

1. **Set AOI**
   Load Burgwald AOI, transform to EPSG:4326, merge to one polygon.
2. **Use project directory structure**
   Sentinel-2 data only in:
   `data/raw/AOI_Burgwald/s2/`
   Training areas in:
   `data/raw/AOI_Burgwald/train_areas_burgwald.gpkg`
3. **Create training areas**
   Digitise polygons inside the AOI, assign attribute `class`.
4. **Run STAC query**
   Use `intersects = AOI`,
   datetime = 2018-05-01 to 2022-12-31,
   send as **POST request**, then call `items_fetch()`.
5. **Select suitable dates**
   Compute per-date scene count + mean cloud cover,
   rank by cloud cover, choose two low-cloud dates (2018 and 2022).
6. **Build predictor stacks**
   For both dates: request raw bands + kNDVI + SAVI + EVI,
   merge into a raster stack, save as:
   `data/processed/pred_stack_2018.tif`
   `data/processed/pred_stack_2022.tif`
7. **Check consistency**
   Ensure identical extent, resolution, and layer order in both stacks.
8. **Document step**